Wilson Pham (Pham Ngoc Son)

Senior Full-Stack & AI Systems Engineer

Ha Noi, Vietnam Global Remote

English: **TOEIC LR 820 / 990** (Professional Working Proficiency)

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PROFESSIONAL SUMMARY

Senior Full-Stack & AI Systems Engineer with over a decade of engineering foundation spanning distributed microservices and production AI platforms. Proven track record in designing cost-effective **LLM guardrails**, authoring multi-engine **GraphRAG** retrieval pipelines, and architecting autonomous coding agents inside hardened **Docker and MCP sandboxes**. Deep expertise building event-driven backend services with **Go and Node.js**, developing reactive **Next.js and React** interfaces, and engineering low-level storage engines with vector indexing. Blends strong systems architecture with robust edge AI workflows to deliver resilient, secure, and **high-throughput enterprise solutions**.

TECHNICAL SKILLS

Full-Stack & Web Architecture: Next.js 14-16 (SSR & App Router), React 18/19, TypeScript, Vite, Tailwind CSS 4, Radix UI, TanStack Query, Zustand, Chrome Extension Manifest V3, WebSockets, REST & GraphQL APIs.

Vibe Coding & Coding Agents: Advanced agentic workflows with Claude Code, Codex, Antigravity (Skills, Agents, MCP servers, Plugins, Rules, Hooks, Sidecars, automated test/lint feedback loops & self-correction).

GraphRAG & Knowledge Retrieval: PathRAG, LightRAG, HippoRAG (Personalized PageRank), Standard GraphRAG, RAPTOR, ToG (Think-on-Graph), GCN, Louvain Clustering, Knowledge Graph Entity-Relationship Extraction.

Database Internals & Storage Engines: CongraphDB (Author), Columnar Storage, HNSW Vector Indexing (297x batch speedup), Write-Ahead Logging (WAL) ACID, `memmap2` Zero-Copy I/O, Neo4j (GDS Certified), PostgreSQL 16 (pgx/v5), Drizzle ORM, Prisma.

Backend & Distributed Microservices: Fastify 4/5, NestJS 10/11, Express, FastAPI (Python), Go (Golang 1.22+, Chi, Watermill), Redis, BullMQ (Delayed Queues & Worker Pools), Apache Kafka / Redpanda, Distributed Sagas, Reverse Proxies, Domain-Driven Design (DDD).

AI/ML & LLM Guardrails: ModernBERT, Llama Guard 3 1B, ONNX Runtime (DirectML/CPU), Vulkan GGUF, Fast-Slow Dual-Process Guardrails, Calibrated Confidence Routing, PyTorch DDP, Discrete Diffusion Models, RLCD, CLM.

Computer Vision & Multimodal VLM: LiquidAI LFM2.5-VL, YOLO26-Pose, YOLO26-Seg, DirectML GPU Acceleration (AMD Radeon), OpenCV, Anatomical Sub-Crop Extraction, Video Management Systems (VMS), Frame Telemetry.

Big Data, Cloud & DevOps: Apache Spark 3.5+, PySpark, Spark MLlib, Kubernetes (k3d), MinIO S3, Medallion Parquet (Bronze/Silver/Gold), Docker Compose (Profile-Gated), Prometheus, Grafana, Loki.

Blockchain and Smart contract: Solidity (0.8.x), OpenZeppelin (ERC20/721, Upgradeable Proxies), Foundry DeFi security audits, GMX-style real-yield multi-token staking, RAsset fractionalization, Circom zk-SNARKs (Groth16), ethers.js v6.

EMPLOYMENT HISTORY & PROFESSIONAL EXPERIENCE

Principal AI Systems & Full-Stack Architect — Independent Consultant / Contract
10/2020 – Present

- Architected and deployed **full-stack web platforms**, **distributed microservices**, and high-performance APIs using **TypeScript**, **Node.js**, **Python**, **PostgreSQL**, **Redis**, and **Docker**.
- Engineered high-throughput **content aggregation** and digital marketing automation pipelines alongside **distributed task orchestration** and **headless browser automation** engines.
- Authored, deployed, and audited production-grade **Solidity smart contracts** (ERC20/721, upgradeable proxies, real-yield staking routers, and asset tokenization); performed DeFi protocol security audits and vulnerability remediation using **Foundry**.

Solutions Architect & Systems Integration Engineer — Lefa Technologies
05/2019 – 10/2020

- Led **technical solution architecture**, partner integrations, and commercial client engagements for enterprise **Smart Building Management Systems (BMS)**, digital transformation, and **IoT platforms**.
- Delivered solutions for landmark enterprise projects: **PVcomBank Headquarters**, **Hoang Cau Building**, and **TH Hospital**. Partnered with **Sun Group**, CPG, Boydens, and PTA.

Technical Solutions & Systems Account Manager — IBS ., JSC (Integrated Building Solutions)
06/2014 – 05/2019

- Directed **technical solutioning**, tender engineering, and partner supplier ecosystems for major **national infrastructure**, **smart city**, and **building automation** projects.
- Key Major Deployments: **Ha Noi – Hai Phong Expressway**, **Deo Ca Tunnel**, **VietinBank Tower Head Office**, Hapulico Complex, TTG Building. Partnered with **Viettel**, **FPT**, Sun Group, Hawee, Sigma, Sao Bac Dau.

Sales Executive & Real Estate Consultant (Collaborator) — CEN LAND
12/2013 – 04/2014

- Real estate commercial brokerage, transaction structuring, client investment portfolio advisory, and contract negotiation for residential and commercial property developments.

Automation Test Engineer — Synergix Technologies
11/2011 – 10/2013

- Architected and implemented **automated regression testing frameworks** for comprehensive multi-tier **Enterprise Resource Planning (ERP)** systems (Finance, CRM, SCM, HR, Manufacturing modules).
- Developed modular, reusable test automation suites using **Java**, **Selenium WebDriver**, **SQL**, **TestNG**, and **CI/CD pipelines**.

Software Test Engineer — FPT Software
04/2010 – 11/2011

- Conducted compiler, toolchain, and IDE quality assurance testing for Freescale Semiconductor's **CodeWarrior Starcore Eclipse 10.0 Development Studio**.
- Designed test plans, **automated regression test cases**, and executed functional test suites within **Eclipse IDE** environments with **CVS** version control.

KEY PROJECTS & AI SYSTEMS ARCHITECTURE

AI, ML and Agent

1. KahnemanGuard — Dual-Process Fast-Slow LLM Guardrail Architecture

Creator & Lead Architect

A high-throughput LLM guardrail system inspired by Daniel Kahneman's *Thinking, Fast and Slow* (VNU-HCM Master's Research), resolving the latency vs. security trade-off through a dual-process cascaded inference pipeline (90.4% compute cost savings , 99.2%–99.6% DSR , 0.0% FPR).

- **Tier 1 (Fast Loop):** Fine-tuned ModernBERT INT8 ONNX encoder classifying safety in 57.3ms CPU P50 (P99: 63.0ms) with a 95.2% fast decision rate.
- **Calibrated Routing (τ = 0.80):** Dual-threshold routing (`τ_safe`, `τ_unsafe`, T=1.9665) resolving high-confidence requests immediately.
- **Tier 2 (Deep Loop):** Vulkan-accelerated GGUF `Llama-Guard-3-1B` triggered conditionally on uncertainty (only 4.8% escalation rate for Dense, 5.2% for Diffusion-RLCD).
- **1,140 Academic Sample Matrix:** Verified on JailbreakBench, ToxicChat, BeaverTails, and XSTest: Dense yields **99.2% DSR** / **0.0% FPR** / Brier 0.0712; Diffusion yields **99.6% DSR**; Diffusion-RLCD yields **99.4% DSR** / 81.2ms P50.

Python | ModernBERT (INT8 ONNX) | Llama-Guard-3-1B (Vulkan) | DirectML | FastAPI | Discrete Diffusion | RLCD | PyTorch DDP

[github: kahneman-guard]

2. CongraphRAG — Unified Multi-Engine GraphRAG Framework

Author & Core Developer

A modular, operator-based GraphRAG execution framework providing native support for 8 GraphRAG retrieval engines backed by ConGraphDB hybrid graph-vector storage.

- **Multi-Engine Orchestration:** Unified execution across PathRAG, LightRAG (dual-level retrieval), HippoRAG (hippocampal associative memory with Personalized PageRank), Standard GraphRAG, RAPTOR (chunk tree summarization), and ToG.
- **Atomic Operator Architecture:** 16+ composable pipeline operators covering entity extraction, edge scoring, and hierarchical community summaries.
- **Quality Assurance:** 290+ automated unit & integration tests verifying graph query determinism and retrieval accuracy across benchmark datasets.

TypeScript | ConGraphDB | PathRAG | LightRAG | HippoRAG | Standard GraphRAG | RAPTOR | ToG | Vitest

[npm: congraph-rag]

3. Auto-Heal Sandbox Agent — Autonomous Code-Repair & Sandboxed Evaluation

Creator & Lead Architect

Local-first autonomous code-repair agent running closed-loop diagnose-patch-execute loops inside hardened, isolated Docker sandboxes powered by edge SLMs.

- **Closed-Loop Healing:** LangGraph state graph evaluating test failures, generating candidate code patches, and iteratively verifying execution with zero human intervention.
- **Hardened Docker Sandbox:** Ephemeral sandbox (`user="nobody"`, 512MB RAM, 1 CPU core, network isolation none, in-memory tarfile injection, 10s execution timeout).
- **Edge SLM Guardrails:** Markdown block regex extraction and 10-line traceback pruning engineered to preserve context windows on local edge SLMs (LiquidAI LFM 2.5 2.6B).

Python 3.11+ | LangGraph | Docker SDK Sandbox | LiquidAI LFM-2.5 (2.6B Q8) | Pytest | Pydantic

[github: auto-heal-sandbox-agent]

4. LexTrade Agent — Autonomous Trade Compliance & Export Control Agent

Lead Agentic AI Engineer

Autonomous enterprise agent for trade compliance screening, export controls classification, and dual-use audits under EU Regulation 2021/821 and the Wassenaar Arrangement.

- **Hermes ReAct Plan-and-Solve Loop:** Multi-turn scratchpad reasoning with parallel multi-syntax tool execution and dynamic duplicate suppression.
- **FastMCP Compliance Tools:** Fuzzy token screening against OFAC SDN & EU Consolidated lists, 6-digit HS tariff classification, and multi-category dual-use threshold verification.
- **Hybrid Episodic Memory:** Sliding-window context paired with persistent SQLite non-repudiation audit journal and trigram semantic precedent retrieval.

Python 3.10+ | FastMCP Protocol | Hermes ReAct | SQLite Audit Journal | Trigram Vector Search | LFM-2.5-2.6B

[github: lex-trade-agent]

5. VN-legal — Vietnam Legal Consultant System

Lead AI Engineer

Intelligent legal consultation platform parsing 689 articles of Vietnam Civil Code into hierarchical clauses and points for semantic retrieval and legal relation discovery.

- **Dual-Phase Retrieval:** Offline vector search via LanceDB (@xenova/transformers ONNX) paired with embedded ConGraphDB for deep legal clause dependency traversal.
- **High-Performance UI:** Sub-150ms interactive Cytoscape/Sigma.js graph rendering coupled with streamed LLM response generation and exact legal citation anchoring.

TypeScript | Fastify | React / Vite | ConGraphDB | LanceDB | ONNX Transformers | Cytoscape / Sigma.js

[github: vn-legal]

6. PersonaTrace — Real-Time AI Video Management System & Persona Tracker

Computer Vision Architect

Production AI VMS ingesting live RTSP/video feeds for real-time person tracking and dual-stage attribute classification (heuristic vision + asynchronous VLM reasoning).

- **GPU-Accelerated Pose Tracking:** DirectML-accelerated `YOLO26n-pose` running at 14-26ms latency on consumer AMD Radeon GPUs.
- **Dual-Stage Attribute Pipeline:** 40-60 FPS real-time heuristic CV paired with background LiquidAI `LFM2.5-VL` reasoning queues over anatomical crops.
- **Real-Time Telemetry:** Binary JPEG frame streaming over WebSockets paired with synchronized JSON bounding box, trajectory, and keypoint coordinates.

Python | FastAPI | YOLO26-Pose | LiquidAI LFM2.5-VL | DirectML | OpenCV | TypeScript | WebSockets

Distributed Platforms, Infrastructure & Storage Engines

7. Deworkers — Distributed Task Orchestration Platform & Multi-Connector Ecosystem

Lead Platform Architect

Enterprise distributed automation platform coordinating multi-tenant background jobs, headless browser workers, and multi-channel API notification connectors.

- **Multi-Tier Job Scheduling:** Redis-backed BullMQ priority queues with exponential backoff, dead-letter routing, and dynamic worker pool autoscaling.
- **Multi-Connector Integration:** Pluggable notification connectors (Telegram, Discord, Slack, Webhooks, Resend Email) with template compilation and rate limiters.
- **Browser Automation Engine:** Playwright headless contexts executing web extraction and synthetic transactions with persistent session replayability.

TypeScript | Node.js | BullMQ | Redis | Playwright | Fastify | Docker | Prometheus

8. InfraHub — Profile-Gated Infrastructure Orchestration Platform

DevOps & Infrastructure Architect

- **Profile-Gated Infrastructure:** Modular Docker Compose topologies isolating lifecycle management across backing services (cache, database, search, storage, and observability).
- **Live-State Introspection API:** Fastify backend continuously inspecting container health, port bindings, resource usage, and storage connectivity with integrated Prometheus/Loki telemetry.

Docker Compose | Fastify | Redis 7 | PostgreSQL | MongoDB | Elasticsearch | Prometheus | Loki | Grafana

9. Event-Driven E-Commerce Microservices

Microservices Architect

Enterprise e-commerce distributed system implementing event-driven Choreography Sagas, polyglot persistence, and high-concurrency order processing.

- **Choreography Saga Engine:** End-to-end distributed transaction orchestration across Order, Payment, Inventory, and Delivery services via Apache Kafka/Redpanda and BullMQ with automated compensating rollbacks.
- **Polyglot Data Architecture:** SQLite/PostgreSQL 15 (Prisma) transactional core, MongoDB 7 high-flexibility catalog, Elasticsearch 8.11 full-text search, Redis L2 caching, and MinIO S3 object storage.
- **Automated CI/CD & K8s Deployment:** GitHub Actions matrix pipelines & Jenkinsfile for automated testing, multi-service Docker container publishing (GHCR), and Kustomize declarative deployments across Staging and Production.

Fastify 4 | Next.js 14 SSR | React 19 / Vite | Apache Kafka / Redpanda | BullMQ Sagas | Prisma ORM | Postgres / Mongo / ES | CI/CD (Actions & Jenkins) | Kubernetes (Kustomize) | Prometheus / Loki

[github: microservices-ecommerce]

10. Domain-Driven Trading Engine

Core Systems Architect

Low-latency institutional securities trading core in Go implementing pure DDD, in-memory matching, pre-trade purchasing power risk checks, and double-entry bookkeeping.

- **In-Memory Matching & O(1) Linked Lists:** Deterministic integer-tick order book (Limit/Market); doubly-linked list price queues ensuring O(1) order operations & cancellations.
- **Pre-Trade Risk & Double-Entry Balancing:** In-memory purchasing power reservations (Price × Qty × (1 + Fee)) with zero overdrafts; atomic journal vouchers satisfying ∑ Debits = ∑ Credits.
- **Event-Driven Settlement & Web Terminal:** Watermill Kafka/Redpanda pipeline with async PostgreSQL 16 batch settlement (`pgx.Batch`), Redis idempotency, and bilingual React terminal with real-time Level 2 Market Depth.

Go (Golang 1.22+) | Hexagonal DDD | In-Memory FIFO Matching | Watermill (Kafka / Redpanda) | Double-Entry Ledger | PostgreSQL 16 (pgx/v5) | Redis 7 | React Terminal

[github: domain-driven-trading]

11. ClimateBigData — Modernized Cloud-Native Analytics with Spark & Kubernetes

Big Data Architect

Modernization of a legacy 5-tier Hadoop MapReduce architecture into a high-throughput, cloud-native Spark on Kubernetes pipeline processing global climate datasets.

- **60x Performance Acceleration:** Reduced processing time from ~26 minutes (Hadoop) to ~26 seconds (Spark), maintaining exact data parity with error delta < 0.00005°C.
- **Medallion Data Lakehouse:** Multi-hop Parquet partitioning (Bronze raw -> Silver cleaned Spark datasets -> Gold analytical aggregates & ML feature sets).
- **Distributed MLlib Forecasting:** Linear & polynomial regression trend forecasting with CO2-temperature correlation models executed on containerized Spark clusters.

PySpark 3.5 | Apache Spark SQL | Spark MLlib | Kubernetes (k3d) | MinIO S3 | Parquet | Streamlit

12. CongraphDB — Embedded Hybrid Graph-Vector Database

Database Architect & Author

An embedded hybrid graph-vector database engine written in Rust with native Node.js FFI bindings, designed for local-first GraphRAG workloads and low-latency traversal.

- **Hybrid Graph-Vector Engine:** Dual-representation storage combining columnar graph property tables with SIMD-accelerated HNSW vector indices delivering 297x batch speedup.
- **Storage Internals & ACID Durability:** Zero-copy memory-mapped I/O (`memmap2`), Write-Ahead Logging (WAL) for crash-safe transactions, and Zstandard block compression.
- **FFI Interoperability:** Zero-overhead Node.js/TypeScript bindings via `napi-rs` with typed JavaScript APIs and built-in Cypher-like traversal operators.

Rust | napi-rs | Node.js 20+ | TypeScript | HNSW Vector Index | Rayon SIMD | WAL (ACID) | memmap2

[npm: congraphdb]

Blockchain & Smart Contract Systems

13. Blockchain Protocols, Real-Yield Staking & Security Collection

Blockchain Protocol Architect

Production-grade decentralized finance protocols, real-world asset (RWA) fractionalization engines, real-yield staking architectures, and offensive/defensive smart contract security audits.

- **Real-Yield Multi-Token Staking:** Architected GMX-style multi-token reward distribution pools featuring reward routers, fee-tracking contracts, and non-dilutive yield accrual.
- **Real-Asset (RAsset) Tokenization:** Engineered fractionalized asset ownership primitives with transparent upgradeable proxies, buyout auctions, and automated yield settlement.
- **Security Audits & Formal Invariants:** Performed comprehensive security audits with Foundry fuzzing, remediating flash-loan reentrancy, oracle manipulation, and ERC20 fee-on-transfer edge cases across 18+ protocol scenarios (Damn Vulnerable DeFi v4 verified).

Solidity 0.8.x | Foundry | Hardhat | OpenZeppelin | Circom zk-SNARKs | snarkjs | ethers.js v6

[github: damn-vulnerable-defi]

CERTIFICATIONS & LANGUAGE PROFICIENCY

TOEIC Listening & Reading: 820 / 990
Score: 820 / 990 (ETS)
Professional Working Proficiency • Global Business & Tech Communication

Claude Code in Action (Anthropic)
Verify Completion • Course Details
Agentic Architectures, Tool Use Protocols, Sandboxing & Multi-Agent Teams

AWS Knowledge: AI Driven Dev Lifecycle
Verify Badge (Credly) • Learning Plan
AI-Driven SDLC, Generative AI Coding, Automation & Responsible AI

AWS Cloud Essentials Badge
Verify Badge (Credly) • Learning Plan
Cloud Computing, Storage, Security, Networking & Distributed Architecture

Neo4j Graph Data Science (GDS)
Verify Certificate • Course Details
Graph Algorithms, Node Embeddings, Community Detection & Link Prediction

Neo4j Certified Professional
Verify Certificate • Course Details
Cypher Query Language, Indexing, Performance Optimization & Graph Models

EDUCATION & ACADEMIC BACKGROUND

University of Information Technology (VNU-HCM)
Bachelor of Science in Artificial Intelligence
AI Safety Guardrails, GNN/GraphRAG, Edge VLM, Columnar Databases, Spark Big Data

Vietnam Maritime University (VMU) (2005 – 2010)
Bachelor of Engineering in Industrial Electrical & Automation Engineering
Graduated with Credit (Khá) • Automated Control Systems, PLC Logic, Electrical Machines

Aptech Computer Education Center (2008 – 2010)
International Higher Diploma in Software Engineering (ACCP i7.1)
International Programmer • Java, C#, C++, SQL, Software Architecture, Enterprise Systems

PTI Business Training School (2010)
Professional Customer Relationship & Negotiation Skills
Client Management, Commercial Negotiation, Enterprise B2B Technical Engagement